UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	DECEMBER 2019
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (1 Question)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

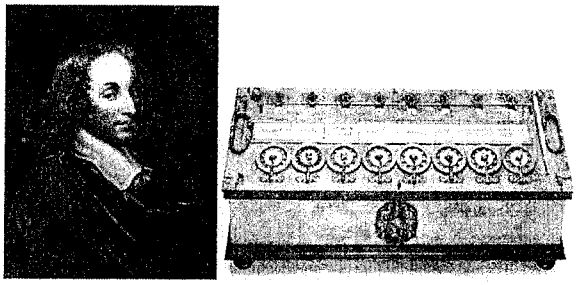
This examination paper consists of 6 printed pages

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PART A (85 MARKS)

QUESTION 1

Question 1 is based on the following Figure 1.



Inventor 'Y' was France's most celebrated mathematician and physicist. In 1642, inventor 'Y' invented and build the first working calculating machine as a means of helping his father to perform tedious tax accounting.

Figure 1: Inventor 'Y' and his calculating machine.

a) Who is inventor 'Y'?

(1 mark)

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

(1 mark)

c) Which programming language is named after inventor 'Y'?

(1 mark)

d) State **TWO (2)** mathematical operations that can be performed by inventor Y's calculating machine.

(2 marks)

QUESTION 2

a) The Von Neumann Architecture was first introduced in which computer generation?

(1 mark)

b) Draw the Von Neumann Architecture.

(2 marks)

c) Central Processing Unit (CPU) is one of the components of a Von Neumann Architecture whereas Control Unit and Arithmetic Logic Unit are the main components of a CPU. What is the main task of a CU?

(2 marks)

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QUESTION 3

a) Calculate $FIFA_{16} - 664_{16}$ using the following techniques. Show the complete workflow.

i) Sign and magnitude

(5 marks)

ii) 2's complement

(5 marks)

QUESTION 4

a) Convert the following values to the respective number bases. Show the complete work flow.

i) 1210.331_8 to hexadecimal

(4 marks)

ii) 295.8125_{10} to octal

(4 marks)

iii) $BACA_{16}$ to binary

(2 marks)

QUESTION 5

a) Differentiate between bit, byte, word and block.

(5 marks)

b) Answer the following questions on IEEE single precision floating point numbers.

i) Convert -151.625_{10} to floating point number in IEEE single precision format in hexadecimal value form.

(5 marks)

ii) Convert $425B900_{16}$ in IEEE single precision floating point number to decimal value form.

(5 marks)

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QUESTION 6

a) Assume that a portion of tags in the cache looks like Table 1. Identify the specific block address location for the following addresses:

i) CC8933
ii) AEEEB5
iii) 9BC287
iv) CD8B78

Table 1: Portion of Tags in Cache

Tag (binary)	Line number (binary)	Address with block			
		00	01	10	11
1010 1101	1110 1100 1101 10	A1	A2	A3	A4
1010 1110	1110 1110 1011 01	B1	B2	B3	B4
1100 1100	1000 1001 0011 00	C1	C2	C3	C4
1100 1101	1000 1011 0111 10	D1	D2	D3	D4
1001 1010	1100 0001 0011 00	E1	E2	E3	E4
1001 1011	1100 0010 1000 01	F1	F2	F3	F4

(4 marks)

b) What is the type of mapping function used in Table 1?

(1 mark)

QUESTION 7

Describe the specific function of different types of registers.

(10 marks)

QUESTION 8

a) State **TWO (2)** different types of bus.

(2 marks)

b) The size of a word is 32 bits and the size of the block is 256 bits. Determine of the following:
i) The byte for word
ii) The byte for block.
iii) Size of the register.
iv) Size of the data bus.

(8 marks)

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QUESTION 9

a) Define interrupt.

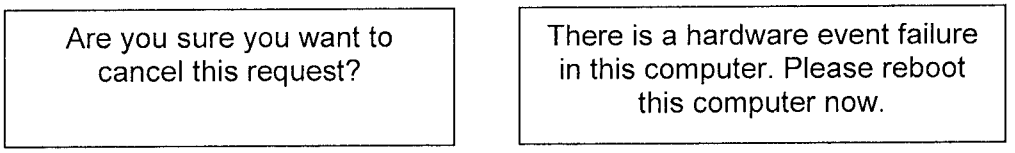
(1 mark)

b) Differentiate between interrupt and direct manipulation access.

(4 marks)

c) Given the following scenario.

Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in *woolworth.com* and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.



(a) (b)

Figure 2: The pop-up messages

i) Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.

(4 marks)

ii) While interrupt routine for Interrupt A is executing, it is interrupted by interrupt B which has a higher priority than interrupt A. What will happen to interrupt routine A?

(1 mark)

QUESTION 10

a) List **THREE (3)** characteristics of Symmetric Multiprocessor (SMP).

(3 marks)

b) Describe **ONE (1)** advantage of Symmetric Multiprocessor (SMP) in terms of *availability*.

(2 marks)

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PART B (15 MARKS)

QUESTION 1

Question 1 is based on Figure 3 and Table 2.

$$y = (A/B^2 - C^2) / (B + C) - A$$

Figure 3: Equation r

Write the appropriate program using correct assembly language notation to compute equation in Figure 3 using **one-address** and **three-address** instructions.

The instructions are given in Table 2. Note that register **A, B** and **C** cannot be modified and register **x, y** and **z** can be used as temporary register. Final answer must be stored in register **y**.

Table 2: Machine instruction

One-address	Three-address
LOAD m	ADD (m, n, o)
STOR m	SUB (m, n, o)
ADD m	MUL (m, n, o)
SUB m	DIV (m, n, o)
MUL m	
DIV m	

(15 marks)

END OF QUESTION PAPER

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